

Syllabus

Academic Year 2026-2027

1. General information							
Course Code	AIP 510						
Course Title	AI Image Processing						
Degree Cycle (Level)/ Major / Relation to curriculum	Master; Applied Artificial Intelligence						
Year, trimester	2nd trimester						
Language of Instruction:	English						
Lecturer(s)/ Instructors/ Instructor Contact Information	Ainur Shekerbek, Assistant Professor, PhD, ainur.shekerbek@astanait.edu.kz, C1 block, room C1.1.336; Wednesday 15:00-17:00 Arman Idrisov, Senior Lecturer and Laboratory Instructor, arman.idrisov@astanait.edu.kz, C1 block, room C1.3.311; by appointment						
Number of Credits	5						
Workload of course components and credits per trimester	ECTS credits	Contact hours			IASS	SIS	Total hours
		Lectures	Practice sessions	Laboratory work			
	5	30	20	10	10	80	150
Prerequisites	Python Programming, Linear Algebra, Probability and Statistics, Machine Learning, and Digital Image Processing.						
Post requisites	Advanced Computer Vision, AI in Medicine and Industry, Research Project, and Master's Thesis.						

2. Goals, objectives and learning outcomes of the course	
1. Course Overview/Description	AI Image Processing is an advanced course on acquisition, enhancement, analysis, understanding, and generation of digital images using classical computer vision and modern artificial intelligence. Students implement algorithms in Python, train deep neural networks, evaluate model quality, and complete a reproducible application-oriented project using a real image dataset.
2. Course Learning Goals	• Build a systematic understanding of image formation, representation, enhancement, segmentation, and analysis. • Connect classical image processing with machine learning and deep learning. • Develop reproducible computer vision experiments using Python. • Select suitable architectures, losses, metrics, and validation strategies. • Evaluate robustness, bias, privacy, explainability, and ethical risks. • Plan, implement, document, and present an applied image processing project.

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3.	Course Learning Outcomes	● Explain sampling, quantization, color models, convolution, filtering, and frequency-domain processing. ● Implement enhancement, restoration, segmentation, feature extraction, and geometric transformations. ● Prepare image datasets and reliable training, validation, and testing pipelines. ● Train neural networks for classification, detection, and semantic segmentation. ● Apply transfer learning, augmentation, regularization, and hyperparameter tuning. ● Evaluate systems with appropriate metrics and qualitative error analysis. ● Assess explainability, robustness, fairness, privacy, and responsible AI risks. ● Present methodology, results, limitations, and recommendations in a technical report.																										
4.	Methods/forms of teaching	● Lectures with visual demonstrations and guided research-paper analysis. ● Laboratory sessions using Python, OpenCV, scikit-learn, and PyTorch. ● Case discussions and peer review of model behavior and errors. ● Independent reading, programming exercises, and dataset analysis. ● Project-based learning with milestones, a report, and a final demonstration.																										
5.	Coursework and Grading Scheme	<table><tr><th>#</th><th>Assessment</th><th>Weight</th><th>Description</th></tr><tr><td>1</td><td>Laboratory work</td><td>20%</td><td>Five reproducible laboratory submissions.</td></tr><tr><td>2</td><td>Individual assignments</td><td>15%</td><td>Two implementation and analysis assignments.</td></tr><tr><td>3</td><td>Midterm assessment</td><td>20%</td><td>Written and practical assessment for weeks 1-7.</td></tr><tr><td>4</td><td>Course project</td><td>15%</td><td>Dataset, model, experiments, report, repository, and presentation.</td></tr><tr><td>5</td><td>Final examination</td><td>30%</td><td>Comprehensive theory and method-selection examination.</td></tr></table>			#	Assessment	Weight	Description	1	Laboratory work	20%	Five reproducible laboratory submissions.	2	Individual assignments	15%	Two implementation and analysis assignments.	3	Midterm assessment	20%	Written and practical assessment for weeks 1-7.	4	Course project	15%	Dataset, model, experiments, report, repository, and presentation.	5	Final examination	30%	Comprehensive theory and method-selection examination.
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6.	Academic Integrity	Students must follow the Rules of Academic Integrity of Astana IT University. Sources, datasets, pretrained models, external code, and generative AI tools must be declared and cited. Plagiarism, fabrication of results, unauthorized cooperation, and submission of another person's work are prohibited.																										

2. Goals, objectives and learning outcomes of the course					
7. Learning resources:	#	Title	Author	Year	Type
	1	Digital Image Processing, 4th Edition ISBN: 9780133356724; Publisher: Pearson	Rafael C. Gonzalez, Richard E. Woods	2018	Main
	2	Computer Vision: Algorithms and Applications, 2nd Edition ISBN: 9783030343712; Publisher: Springer	Richard Szeliski	2022	Main
	3	Deep Learning ISBN: 9780262035613; Publisher: MIT Press	Ian Goodfellow, Yoshua Bengio, Aaron Courville	2016	Main
	4	Dive into Deep Learning ISBN: 9781009389433; Publisher: Cambridge University Press	Aston Zhang, Zachary Lipton, Mu Li, Alexander Smola	2023	Additional
	5	Deep Learning for Vision Systems ISBN: 9781617296192; Publisher: Manning	Mohamed Elgendy	2020	Additional
	6	Компьютерное зрение. Современный подход ISBN: 9785970609573; Publisher: DMK Press	Richard Szeliski	2023	Additional
	7	Жасанды интеллект және кескіндерді өңдеу материалдары Publisher: AITU	Astana IT University	2026	Additional
8. Technology employed:	• Moodle and the university corporate email service. • Python 3.12, JupyterLab, NumPy, OpenCV, scikit-image, scikit-learn, and PyTorch. • Git and GitHub/GitLab for version control and reproducibility. • GPU-enabled workstations or approved cloud notebook environments. • MegaPro university library catalogue.				
9. Course Policies	Attendance: Attendance and active participation are compulsory. Students normally need at least 70% attendance to be admitted to the final examination. Late submissions: Work must be submitted through Moodle by the stated deadline. Extensions require prior approval and supporting evidence. Attestation I and II: Students must satisfy attestation thresholds and complete mandatory laboratory work. Examination procedures follow current university policy. Classroom behavior: Students must communicate respectfully, follow laboratory safety and cybersecurity requirements, and use devices only for learning activities. Communication: Announcements are published in Moodle. Corporate-email questions are normally answered within two business days. Consultations are available during office hours. Other policies: Personal or sensitive datasets require documented authorization. Students must follow licenses and must not upload restricted data to unapproved services.				
10. Course Schedule	See the Course Schedule section below.				
11. Syllabus Inclusion Statements	The course supports an inclusive and respectful learning environment. Students may request reasonable accommodations through university procedures while preserving the intended learning outcomes of the course.				

3.1 Abbreviations		
#	Abbreviation	Meaning
1	ECTS	European Credit Transfer and Accumulation System
2	IASS	Independent academic study supervised by the instructor
3	SIS	Student independent study
4	CNN	Convolutional neural network
5	IoU	Intersection over Union
6	mAP	Mean Average Precision
7	ViT	Vision Transformer

3.2 Course Schedule					
Week	Topic	Lecture / Theory	Practice / Lab	Independent work	Assessment / Resources
1	Digital image foundations	Image formation, sampling, quantization, color spaces, and representations.	Environment setup, image I/O, color conversion, and histogram analysis.	Read chapters 1-2 and analyze a selected dataset.	Diagnostic quiz and laboratory preparation.
2	Spatial image enhancement	Point operations, convolution, smoothing, sharpening, and edge-preserving filters.	Implement spatial filters and compare image quality.	Exercises on kernels, border handling, noise, and contrast.	Laboratory 1: Image enhancement.
3	Frequency-domain processing	Fourier transform, spectra, low-pass and high-pass filtering, and restoration.	Implement frequency-domain filtering in Python.	Compare spatial and frequency-domain approaches.	Assignment 1 released.
4	Segmentation and morphology	Thresholding, regions, watershed, morphology, and connected components.	Segment objects and evaluate binary masks.	Review segmentation metrics and annotate validation data.	Laboratory 2: Segmentation.
5	Features and classical recognition	Corners, descriptors, feature matching, verification, and classifiers.	Build a feature-matching and classification pipeline.	Prepare an error analysis for descriptor matching.	Assignment 1 due.
6	Neural network foundations	Backpropagation, optimization, regularization, and loss functions.	Build a baseline network and analyze training curves.	Complete PyTorch tensor and optimization exercises.	Laboratory 3: Training pipeline.
7	Convolutional neural networks	CNN architecture, receptive fields, normalization, residual connections, and capacity.	Train and evaluate a multiclass CNN.	Compare architectures and document the protocol.	Midterm review and project proposal due.
8	Transfer learning and augmentation	Pretrained models, fine-tuning, augmentation, imbalance, and domain shift.	Fine-tune a model and conduct an ablation study.	Prepare project data, licensing notes, and data splits.	Midterm assessment; Laboratory 4.

3.2 Course Schedule					
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9	Object detection	One-stage and two-stage detectors, IoU, precision-recall, and mAP.	Evaluate detector predictions and analyze errors.	Read a detection paper and prepare project milestone 1.	Assignment 2 released.
10	Semantic and instance segmentation	Encoder-decoder models, U-Net, mask prediction, losses, and metrics.	Train a compact segmentation model.	Prepare qualitative and quantitative comparisons.	Laboratory 5: Semantic segmentation.
11	Generative models for images	Autoencoders, VAEs, GANs, diffusion models, and evaluation.	Experiment with a pretrained generative model.	Review synthetic image quality, limitations, and misuse.	Assignment 2 due.
12	Attention and vision transformers	Self-attention, patch embeddings, positional encoding, and ViT.	Fine-tune a compact transformer and compare with a CNN.	Read the ViT paper and prepare project milestone 2.	Project checkpoint and peer review.
13	Explainability and responsible AI	Attribution, adversarial examples, uncertainty, bias, privacy, and model cards.	Apply explainability tools and robustness tests.	Draft responsible AI and limitations sections.	Responsible AI case analysis.
14	Deployment and reproducibility	Inference optimization, export, tracking, testing, documentation, and monitoring.	Package the inference pipeline and verify reproducibility.	Complete report, repository, and presentation.	Project readiness review.
15	Project presentations	Integration of methods, evaluation design, communication, and future directions.	Demonstrate the system and defend design decisions.	Submit report, code, model card, and reflection.	Final project presentation and submission.

Additional Sections	
Assessment Requirements	Every graded submission must include reproducible code, declared dependencies, methodology, results, error analysis, and references.

Custom Sections	
Consultation and Student Support	University academic and wellbeing services remain available throughout the trimester. Consultations Consultations are available during office hours and by appointment. Student support Students should request support early for mathematics, programming, accessibility, or project planning.